

Course Code	21CSE524T	Course Name	COMPUTER VISION TECHNIQUES	Course Category	E	PROFESSIONAL ELECTIVE	L	T	P	C
							2	1	0	3

Pre-requisite Courses	Nil	Co-requisite Courses	Nil	Progressive Courses	Nil
Course Offering Department	School of Computing	Data Book / Codes / Standards	Nil		

Course Learning Rationale (CLR):		The purpose of learning this course is to:												Program Specific Outcomes		
CLR-1:	recognize and describe both the theoretical and practical aspects of computing with images.connect issues from computer vision to human vision	1	2	3	4	5	6	7	8	9	10	11	12			
CLR-2:	describe the foundation of image formation and image analysis, Understand the basics of 2D and 3D Computer Vision	Engineering Knowledge	Problem Analysis	Design/development of solutions	Conduct investigations of complex problems	Modern Tool Usage	The engineer and society	Environment & Sustainability	Ethics	Individual & Team Work	Communication	Project Mgt. & Finance	Life Long Learning	PSO-1	PSO-2	PSO-3
CLR-3:	become familiar with the major technical approaches involved in computer vision, describe various methods used for registration, alignment, and matching in images															
CLR-4:	get an exposure to advanced concepts leading to object and scene categorization from images															
CLR-5:	build computer vision applications															
Course Outcomes (CO):		At the end of this course, learners will be able to:														
CO-1:	understand computer vision including fundamentals of image formation	-	-	-	-	-	-	-	-	-	-	-	-	2	-	-
CO-2:	learn about image formation	-	-	-	-	-	-	-	-	-	-	-	-	-	-	2
CO-3:	apply image processing techniques	-	-	-	-	-	-	-	-	-	-	-	-	1	-	1
CO-4:	calibrate computational photography	-	-	-	-	-	-	-	-	-	-	-	-	-	1	-
CO-5:	learn about Image rendering	-	-	-	-	-	-	-	-	-	-	-	-	-	-	1

Unit-1 - Introduction to Computer Vision	9 Hour
Review of image processing techniques – classical filtering operations – thresholding techniques – edge detection techniques – corner and interest point detection. T1: Implementation of 2D and 3D Projections T2: Build a color constancy algorithm that uses the assumption that the spatial average of reflectance is constant. Use finite-dimensional linear models.	
Unit-2 – Image Formation	9 Hour
Points and patches-An Introduction - Binary shape analysis – connectedness – object labeling and counting – size filtering – distance functions – skeletons and thinning – deformable shape analysis – boundary tracking procedures – active contours – shape models and shape recognition - Feature detectors -Feature descriptors, Matching - Feature tracking - Edge detection - Edge linking - Successive approximation - Hough transforms -Hough Transform (HT) for line detection – foot-of-normal method – line localization – line fitting – RANSAC for straight line detection – HT based circular object detection – accurate center location – speed problem – ellipse detection – Case study: Human Iris location – hole detection – generalized Hough Transform (GHT) – spatial matched filtering – GHT for ellipse detection – object location – GHT for feature collation. Vanishing points T3: Implementation Two-dimensional Fourier transforms, Wiener filtering T4: Write a program that produces a Gaussian pyramid from an image	

Unit-3 – Image Processing Techniques	9 Hour
Active contours - Snakes -Dynamic snakes and Condensation - Scissors, Level Sets - Split and merge - Mean shift and mode finding - Normalized cuts - Graph cuts and energy-based methods - 2D and 3D feature-based alignment - Pose estimation T5: Obtain an implementation of Canny's edge detector T6: Implement a Hough transform based line finder	
Unit-4 – Computational Photography	9 Hour
Triangulation - Two-frame structure from motion - Projective reconstruction - Self-calibration - Perspective and projective factorization - Bundle adjustment - Exploiting sparsity - Constrained structure and motion - Hierarchical motion estimation - Fourier-based alignment - Incremental refinement T7: Count lines with an HT line finder. How well does it work? T8: Implement a mean shift segmenter.	
Unit-5 – Image Rendering	9 Hour
Motion models - Planar perspective motion - Rotational panoramas - Gap closing - Cylindrical and spherical coordinates - Bundle adjustment - Parallax removal - Recognizing panoramas – Compositing T9: Case Study T10: Case Study	

Learning Resources	1. Richard Szeliski, "Computer Vision: Algorithms and Applications", Springer, 2010. 2. Forsyth/Ponce, "Computer Vision: A Modern Approach", Pearson Education India, 2nd edition (2015) 3. S. Nagabhushana, "Computer Vision and Image Processing", New Age International Pvt Ltd; First edition (2005)	4. Rafael C. Gonzalez "Digital Image Processing", Pearson Education; Fourth edition (2018)
---------------------------	--	--

Learning Assessment							
	Bloom's Level of Thinking	Continuous Learning Assessment (CLA)				Summative Final Examination (40% weightage)	
		Formative CLA-1 Average of unit test (50%)		Life-Long Learning CLA-2 (10%)			
		Theory	Practice	Theory	Practice	Theory	Practice
Level 1	Remember	15%	-	15%	-	15%	-
Level 2	Understand	25%	-	20%	-	25%	-
Level 3	Apply	30%	-	25%	-	30%	-
Level 4	Analyze	30%	-	25%	-	30%	-
Level 5	Evaluate	-	-	10%	-	-	-
Level 6	Create	-	-	5%	-	-	-
	Total	100 %		100 %		100 %	

Course Designers			
Experts from Industry		Experts from Higher Technical Institutions	Internal Experts
1. Mr. N. Mohanraj, Software Engineer 2, Pay Pal Inc, Chennai		1. Dr. A.P. Shanthy, CEG Campus Anna University	1. Dr. T. Veeramakali, SRMIST